

Homework 3

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*All of my homework is available at <https://github.com/lacampanella233/Homework.git>.

Problem 1 (3C-1). Suppose $T \in \mathcal{L}(V, W)$. Show that w.r.t each choice of basis of V and W , the matrix of T has at least $\dim \operatorname{im} T$ nonzero entries.

Proof. Pick two basis in V and W , denoted by $\mathcal{B} = \{b_1, \dots, b_n\}$ and $\mathcal{C} = \{c_1, \dots, c_m\}$ respectively. Set $M := \mathcal{M}(T; \mathcal{B}, \mathcal{C})$ being the matrix of T , and $d := \dim \operatorname{im} T$.

Suppose that M has at most $d - 1$ entries. Then M has at least $n - d + 1$ columns with all-zero entries, which indicates that there are at least $n - d + 1$ zero vectors among the n vectors $Mb_1, \dots, Mb_n$. Then

$$\dim \operatorname{im} T = \dim \operatorname{span} \{Mb_i : 1 \leq i \leq n\} \leq n - (n - d + 1) = d - 1.$$

Hence, we get a contradiction, and the matrix M has at least d nonzero entries. $\square$

Problem 2 (3C-2). Suppose $T \in \mathcal{L}(V, W)$, where V and W are finite-dimensional and nonzero. Prove that $\dim \operatorname{im} T = 1$ if and only if there exist a basis of V and a basis of W such that, with respect to these bases, all entries of $\mathcal{M}(T)$ equal 1.

Proof. (a) We will prove " $\Rightarrow$ " first. Since $\dim \operatorname{im} T = 1$, there exists a nonzero vector $w_0 \in W$ such that $\operatorname{im} T = \operatorname{span} \{w_0\}$. We can extend w_0 to a basis of W , denoted by $\mathcal{C}' = \{w_0, w_1, \dots, w_{m-1}\}$. Let $w_m = w_0 - w_1 - \dots - w_{m-1}$, it's easy to verify that $\mathcal{C} := \{w_1, \dots, w_m\}$ is also a basis of W .

Next, pick a basis of V in $V - \ker T$ (this is a n -dimensional vector space minus an at most $(n - 1)$ -dimensional subspace, in which there do exist a basis of V), denoted by $\mathcal{B}' = \{v'_1, \dots, v'_n\}$. Since $0 \neq Tv'_i \in \operatorname{im} T = \operatorname{span} \{w_0\}$, there exists a scalar a_i such that $Tv'_i = a_i w_0$. Then set

$$\mathcal{B} := \{a_1^{-1}v'_1, \dots, a_n^{-1}v'_n\}.$$

It's easy to verify that $\mathcal{B}$ is also a basis of V , and for every $v \in \mathcal{B}$, $Tv = w_0 = w_1 + \dots + w_m$. Hence w.r.t the bases $\mathcal{B}$ and $\mathcal{C}$, the matrix of T has all entries equal to 1.

(b) We will prove " $\Leftarrow$ " now. Suppose that there exist a basis of V and a basis of W such that, with respect to these bases, all entries of $\mathcal{M}(T)$ equal 1. Then for every vector $v \in V$, we can write

$v = \sum_{i=1}^n a_i v_i$, where $v_1, \dots, v_n$ are the basis of V . Then

$$Tv = T\left(\sum_{i=1}^n a_i v_i\right) = \sum_{i=1}^n a_i T v_i = \sum_{i=1}^n a_i (w_1 + \dots + w_m) = (a_1 + \dots + a_n)(w_1 + \dots + w_m),$$

where $\{w_i : 1 \leq i \leq m\}$ is the selected basis of W . Hence, $\text{im } T \subseteq \text{span}\{w_1 + \dots + w_m\}$, which indicates that $\dim \text{im } T \leq 1$. Since $T \neq 0$, we have $\dim \text{im } T = 1$. $\square$

Problem 3 (3C-6). Suppose $v_1, \dots, v_m$ is a basis of V and W is finite-dimensional. Suppose $T \in \mathcal{L}(V, W)$. Prove that there exists a basis $w_1, \dots, w_n$ of W such that all entries in the first column of $\mathcal{M}(T)$ [with respect to the bases $v_1, \dots, v_m$ and $w_1, \dots, w_n$] are 0 except for possibly a 1 in the first row, first column.

Proof. (1) If $Tv_1 = 0$, then we can pick an arbitrary basis of W and the conclusion holds.

(2) If $Tv_1 \neq 0$, we can extend Tv_1 to a basis of W , denoted by $w_1 = Tv_1, w_2, \dots, w_n$. Then w.r.t the bases $v_1, \dots, v_m$ and $w_1, \dots, w_n$, the matrix of T has all entries in the first column equal to 0 except for a 1 in the first row, first column. $\square$

Problem 4 (3C-9). Suppose $a = (a_1 \ \dots \ a_n)$ is a 1-by- n matrix and B is an n -by- p matrix. Prove that

$$aB = a_1 B_{1,\cdot} + \dots + a_n B_{n,\cdot}.$$

In other words, show that aB is a linear combination of the rows of B , with the scalars that multiply the rows coming from a .

Proof. Let $B_{i,\cdot}$ be the i -th row of B . Then by the definition of block matrix multiplication, we have

$$aB = (a_1 \ \dots \ a_n) \begin{pmatrix} B_{1,\cdot} \\ \vdots \\ B_{n,\cdot} \end{pmatrix} = \sum_{i=1}^n a_i B_{i,\cdot}.$$

$\square$

Problem 5 (3C-12). Prove that matrix multiplication is associative. In other words, suppose A , B , and C are matrices whose sizes are such that $(AB)C$ makes sense. Explain why $A(BC)$ makes sense and prove that

$$(AB)C = A(BC).$$

Proof. Suppose $A \in \mathbb{F}^{m \times n}$ and $B \in \mathbb{F}^{n \times k}$. Since $(AB)C$ make sense, $C \in \mathbb{F}^{k \times l}$ for some positive integer l . Then BC is an $n \times l$ matrix, and $A(BC)$ makes sense.

Denote the entries of A , B , and C by A_{ij} , B_{ij} , and C_{ij} respectively. Then

$$((AB)C)_{s,t} = \sum_{j=1}^k \left(\sum_{i=1}^n A_{si} B_{ij} \right) C_{jt}$$

and

$$(A(BC))_{s,t} = \sum_{i=1}^n A_{si} \left(\sum_{j=1}^k B_{ij} C_{jt} \right).$$

By the associative property of scalar multiplication and addition, these two expressions are equal. Thus, $(AB)C = A(BC)$. $\square$

Problem 6 (3C-16). Suppose A is an m -by- n matrix with $A \neq 0$. Prove that the rank of A is 1 if and only if there exist $(c_1, \dots, c_m) \in \mathbb{F}^m$ and $(d_1, \dots, d_n) \in \mathbb{F}^n$ such that $A_{j,k} = c_j d_k$ for every $j = 1, \dots, m$ and every $k = 1, \dots, n$.

Proof. (a) We will prove " $\Rightarrow$ " first. Since the rank of A is 1, all the column vectors of A (i.e. $A_{\cdot,i}$ for $1 \leq i \leq n$) lies in a one-dimensional subspace of $\mathbb{F}^m$. Hence, there exists a nonzero vector $c = (c_1, \dots, c_m) \in \mathbb{F}^m$ such that for every $1 \leq i \leq n$, there exists a scalar d_i such that $A_{\cdot,i} = d_i c$. Then we have $A_{j,k} = c_j d_k$ for every $j = 1, \dots, m$ and every $k = 1, \dots, n$.

(b) We will prove " $\Leftarrow$ " now. Suppose there exist $(c_1, \dots, c_m) \in \mathbb{F}^m$ and $(d_1, \dots, d_n) \in \mathbb{F}^n$ such that $A_{j,k} = c_j d_k$ for every $j = 1, \dots, m$ and every $k = 1, \dots, n$. Denote $c = (c_1, \dots, c_m)$. Then, by the definition of rank, we have

$$\text{rank } A = \dim \text{span}\{A_{\cdot,1}, \dots, A_{\cdot,n}\} = \dim \text{span}\{d_1 c, \dots, d_n c\} = 1.$$

$\square$

Problem 7 (3C-17). Suppose $T \in \mathcal{L}(V)$, and $u_1, \dots, u_n$ and $v_1, \dots, v_n$ are bases of V . Prove that the following are equivalent:

- (a) T is injective.
- (b) The columns of $\mathcal{M}(T)$ are linearly independent in $\mathbb{F}^{n,1}$.
- (c) The columns of $\mathcal{M}(T)$ span $\mathbb{F}^{n,1}$.
- (d) The rows of $\mathcal{M}(T)$ span $\mathbb{F}^{1,n}$.
- (e) The rows of $\mathcal{M}(T)$ are linearly independent in $\mathbb{F}^{1,n}$.

Here $\mathcal{M}(T)$ means $\mathcal{M}(T, (u_1, \dots, u_n), (v_1, \dots, v_n))$.

Proof. Denote $M := \mathcal{M}(T) = (c_1, \dots, c_n) = (r_1^T, \dots, r_n^T)^T$, where $c_i \in \mathbb{F}^{n \times 1}$ and $r_i \in \mathbb{F}^{1 \times n}$.

(a) $\Rightarrow$ (b): if $c_1, \dots, c_n$ are linearly dependent, then there exists not-all-zero scalars $a_1, \dots, a_n$ such that $a_1 c_1 + \dots + a_n c_n = 0$, which means $T(a_1 u_1 + \dots + a_n u_n) = 0$, contradicting to (a).

(b) $\Rightarrow$ (c): $c_1, \dots, c_n \in \mathbb{F}^{n \times 1}$ are linearly independent, then $\{c_1, \dots, c_n\}$ is a basis of $\mathbb{F}^{n \times 1}$, which implies the conclusion.

(c) $\Rightarrow$ (a): (c) means that T is surjective, hence $V = \text{im } T = \text{span}\{Tu_1, \dots, Tu_n\}$. Since $\dim V = n$, $\{Tu_1, \dots, Tu_n\}$ is a basis of V . Hence, T is injective.

(c) $\Leftrightarrow$ (d): (c) $\Leftrightarrow$ Column rank of M equals $n \Leftrightarrow$ Row rank of M equals $n \Leftrightarrow$ (d).

(d) $\Leftrightarrow$ (e): (d) $\Leftrightarrow \dim \text{span}\{r_1, \dots, r_n\} = n$. Since $\dim \mathbb{F}^{1 \times n} = n$, that is equivalent to $r_1, \dots, r_n$ are linearly independent. $\square$

Problem 8 (3D-3). Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that the following are equivalent:

- (a) T is invertible.
- (b) $Tv_1, \dots, Tv_n$ is a basis of V for every basis $v_1, \dots, v_n$ of V .
- (c) $Tv_1, \dots, Tv_n$ is a basis of V for some basis $v_1, \dots, v_n$ of V .

Proof. (a) $\implies$ (b): Since T is invertible, T^{-1} exists. For every basis $v_1, \dots, v_n$ of V , if $Tv_1, \dots, Tv_n$ is not a basis of V , then by definition, there exists a vector $v_0 = \sum_{i=1}^n a_i Tv_i = \sum_{i=1}^n b_i Tv_i$, where $a_i, b_i \in \mathbb{F}$ with $(a_1, \dots, a_n) \neq (b_1, \dots, b_n)$. Hence

$$T^{-1}v_0 = \sum_{i=1}^n a_i v_i \neq \sum_{i=1}^n b_i v_i = T^{-1}v_0,$$

which contradicts the linearly independence of $v_1, \dots, v_n$.

(b) $\implies$ (c): trivial.

(c) $\implies$ (a): Since $Tv_1, \dots, Tv_n$ is a basis of V , we can define a linear map $S \in \mathcal{L}(V)$ by setting $S(Tv_i) = v_i$ for every $1 \leq i \leq n$. Then for every vector $v = \sum_{i=1}^n a_i v_i \in V$, we have

$$STv = S\left(\sum_{i=1}^n a_i Tv_i\right) = \sum_{i=1}^n a_i S(Tv_i) = \sum_{i=1}^n a_i v_i = v.$$

Hence, $ST = I_V$. Similarly, we can verify that $TS = I_V$. Thus, T is invertible. $\square$

Problem 9 (3D-6). Suppose W is finite-dimensional and $S, T \in \mathcal{L}(V, W)$. Prove that $\ker S = \ker T$ if and only if there exists an invertible $E \in \mathcal{L}(W)$ such that $S = ET$.

Proof. (a) We will prove " $\implies$ " first. Denote $K = \ker S = \ker T$. K is a subspace of V , hence there exists a nonzero subspace U of V such that $V = K \oplus U$. Thus we can find a basis of U , denoted as $\{u_1, \dots, u_m\}$.

It's easy to verify that $\mathcal{S} := \{Su_i : 1 \leq i \leq m\}$ and $\mathcal{T} := \{Tu_i : 1 \leq i \leq m\}$ are also linearly independent. Then we can extend $\mathcal{S}$ and $\mathcal{T}$ to basis of V :

$$\mathcal{S}' := \{u'_i = Su_i : 1 \leq i \leq m\} \cup \{u'_i : m < i \leq n\}, \quad \mathcal{T}' := \{u''_i = Tu_i : 1 \leq i \leq m\} \cup \{u''_i : m < i \leq n\}.$$

Define E as follow: $Eu''_i = u'_i$. Then, it's easy to verify that $S = ET$ holds in K and U , consequently in V .

(b) Next, we will prove " $\impliedby$ ". Using the invertibility of E , we have

$$Tv = 0 \iff ETv = 0 \iff Sv = 0.$$

$\square$

Problem 10 (3D-10). Suppose V and W are finite-dimensional and U is a subspace of V . Let $\mathcal{E} = \{T \in \mathcal{L}(V, W) : U \subseteq \ker T\}$.

- (a) Show that $\mathcal{E}$ is a subspace of $\mathcal{L}(V, W)$.
- (b) Find a formula for $\dim \mathcal{E}$ in terms of $\dim V$, $\dim W$, and $\dim U$.

Solution.

- (a) We check the subspace criteria.
 - (1) The zero map belongs to $\mathcal{E}$ because $\ker 0 = V$ contains U .
 - (2) If $S, T \in \mathcal{E}$ then $U \subseteq \ker S \cap \ker T$, so $(S + T)u = Su + Tu = 0$ for every $u \in U$. Hence $U \subseteq \ker(S + T)$.
 - (3) If $S \in \mathcal{E}$ and $a \in \mathbb{F}$ then $(aS)u = a(Su) = 0$ for every $u \in U$, giving $aS \in \mathcal{E}$.
- (b) We will show that $\mathcal{E} \cong \mathcal{L}(V/U, W)$, and thus

$$\dim \mathcal{E} = \dim(V/U) \cdot \dim W = (\dim V - \dim U) \dim W.$$

Define

$$f : \mathcal{E} \rightarrow \mathcal{L}(V/U, W), \quad T \mapsto (T' : V/U \rightarrow W, \quad v + U \mapsto T(v)).$$

$f(T)$ is well defined, since if $v_1 + U = v_2 + U$, then $v_1 - v_2 \in U \subseteq \ker T$, which means $T(v_1) = T(v_2)$. It's easy to verify that f is linear. Next, we will show that f is bijective.

- (1) f is injective. Suppose $f(T_1) = f(T_2)$, then for any $v \in V$,

$$T_1(v) = (fT_1)(v + U) = (fT_2)(v + U) = T_2(v).$$

- (2) f is surjective. For any $T' \in \mathcal{L}(V/U, W)$, define $T : V \rightarrow W$ by $Tv = T'(v + U)$. It's easy to verify that $T \in \mathcal{E}$ and $f(T) = T'$.

Hence, f is an isomorphism, and the conclusion follows.

Problem 11 (3D-11). Suppose V is finite-dimensional and $S, T \in \mathcal{L}(V)$. Prove that ST is invertible if and only if both S and T are invertible.

Proof. On the one hand, if S and T are invertible, then S^{-1} and T^{-1} exist. Hence, set $(ST)^{-1} = T^{-1}S^{-1}$. Then $(ST)^{-1}(ST) = T^{-1}S^{-1}ST = T^{-1}T = I$ and $(ST)(ST)^{-1} = STT^{-1}S^{-1} = SS^{-1} = I$, which means ST is invertible.

On the other hand, if ST is invertible, then $STv = 0 \iff v = 0$. If T is not invertible, then $\ker T \neq \{0\}$, which shows there exists $v \neq 0$, such that $Tv = 0$. Then $STv = 0$, which contradicts to the invertibility of ST . Hence, T is invertible. Similarly, if S is not invertible, then there exists $u \neq 0$ such that $Su = 0$. Notice that $T^{-1}u \neq 0$ since T is invertible, then $ST(T^{-1}u) = Su = 0$, which contradicts to the invertibility of ST . Hence, S is invertible. $\square$

Problem 12 (3D-12). Suppose V is finite-dimensional and $S, T, U \in \mathcal{L}(V)$ satisfy $STU = I$. Show that T is invertible and that $T^{-1} = US$.

Proof. By the previous problem, S, T, U are all invertible. Then

$$T^{-1} = UU^{-1}T^{-1}S^{-1}S = U(STU)^{-1}S = UIS = US.$$

□

Problem 13 (3D-13). Show that the result in Exercise 3D-12 can fail without the hypothesis that V is finite-dimensional.

Proof. Take $V = \mathbb{F}^\infty$, and

$$S(a_1, a_2, a_3, \dots) = (a_3, \dots), \quad T = U, \quad U(a_1, a_2, \dots) = (0, a_1, a_2, \dots).$$

It's easy to verify that $STU = I$, but T is not invertible since $\text{im } T \neq \mathbb{F}^\infty$.

□

Problem 14 (3D-17). Suppose V is finite-dimensional and $S \in \mathcal{L}(V)$. Define $\mathcal{A} \in \mathcal{L}(\mathcal{L}(V))$ by $\mathcal{A}(T) = ST$ for $T \in \mathcal{L}(V)$.

- (a) Show that $\dim \ker \mathcal{A} = (\dim V)(\dim \ker S)$.
- (b) Show that $\dim \text{im } \mathcal{A} = (\dim V)(\dim \text{im } S)$.

Proof. (a) Notice that $\ker \mathcal{A} = \{T \in \mathcal{L}(V) : \text{im } T \subset \ker S\}$. Pick a basis $\mathcal{B}' := \{b_1, \dots, b_m\}$ of $\ker S$, and extend it to a basis $\mathcal{B} := \{b_1, \dots, b_n\}$ of V . Then it's obvious to verify that the mn linear maps

$$T_{i_1, \dots, i_n} : V \rightarrow \ker S, \quad b_j \mapsto b_{i_j} \quad (1 \leq j \leq n), \quad 1 \leq i_1, \dots, i_n \leq m$$

form a basis of $\ker \mathcal{A}$. Hence, $\dim \ker \mathcal{A} = mn = (\dim V)(\dim \ker S)$.

(b) We will show that $\text{im } \mathcal{A} = \{T' \in \mathcal{L}(V) : \text{im } T' \subset \text{im } S\}$. On the one hand, for any $T \in \mathcal{L}(V)$, $\text{im } ST \subset \text{im } S$, hence $\text{im } \mathcal{A} \subset \{T' \in \mathcal{L}(V) : \text{im } T' \subset \text{im } S\}$. On the other hand, For any $U \subset \text{im } S$, it's straight to verify that $S^{-1}(U) := \{v \in V : Sv \in U\}$ is a subspace of V . Construct $T \in \mathcal{L}(V)$ such that $\text{im } T = S^{-1}(U)$, thus $\text{im } ST = U$. Therefore $\text{im } \mathcal{A} \supset \{T' : \text{im } T' \subset \text{im } S\}$, and consequently our statement has been proved.

By the totally same argument as that in (a), we can prove that $\dim\{T' : \text{im } T' \subset \text{im } S\} = (\dim V)(\dim \text{im } S)$.

□

Problem 15 (3D-23). Suppose that $u_1, \dots, u_n$ and $v_1, \dots, v_n$ are bases of V . Let $T \in \mathcal{L}(V)$ be such that $Tv_k = u_k$ for each $k = 1, \dots, n$. Prove that

$$\mathcal{M}(T, (v_1, \dots, v_n)) = \mathcal{M}(I, (u_1, \dots, u_n), (v_1, \dots, v_n)).$$

Proof. Denote $\mathcal{V} := \{v_i : 1 \leq i \leq n\}$ and $\mathcal{U} := \{u_i : 1 \leq i \leq n\}$. Then by the definition,

$$\mathcal{M}(T, \mathcal{V}) = ([Tv_1]_{\mathcal{V}}, \dots, [Tv_n]_{\mathcal{V}}) = ([u_1]_{\mathcal{V}}, \dots, [u_n]_{\mathcal{V}}) = \mathcal{M}(I, \mathcal{U}, \mathcal{V}).$$

□